

Last time

Defⁿ (Milnor). $K_2(R) = \ker(\text{St}(R) \rightarrow E(R))$

- $\text{St}(R)$ is generated by $e_{ij}(r)$ / "obvious" relations

so $K_2(R) =$ "extra relations in $E(R)$ "

- Steinberg showed

$$0 \rightarrow K_2(R) \rightarrow \text{St}(R) \rightarrow E(R) \rightarrow 0$$

is a universal central extension of $E(R)$

general theory $\Rightarrow K_2(R) = H_2(E(R), \mathbb{Z})$

Historical timeline

1950 - early 1960s : K_0, K_1 become established

1962 : Steinberg's work in group theory

~1967 : Milnor became interested and defined $K_2 = \ker(\text{St} \rightarrow E)$

$$\text{Ex} / K_2(\mathbb{Z}) = \mathbb{Z}/2\mathbb{Z} = \langle w^4 \rangle$$

$$\text{where } w = e_{12}(1) e_{21}(-1) e_{12}(1)$$

$$(\text{In } E(R), w \text{ evaluates to } \begin{pmatrix} 0 & 1 \\ -1 & 0 \end{pmatrix})$$

$$\text{so } w^4 = 1 \text{ in } E(R)$$

But w^4 cannot be expressed in terms of Steinberg relations)

$$\text{Ex} / K_2(k[t_1, \dots, t_n]) = K_2(k) \quad (A' \text{-inv})$$

$$K_2(k[t, t^{-1}]) = K_2(k) \oplus K_1(k) = K_2(k) \oplus k^\times$$

(Bass - Heller - Swan)

Thm (Matsumoto).

k field

$$\Rightarrow K_2(k) \cong \frac{k^\times \otimes k^\times}{\langle a \otimes (1-a) : a \neq 0, 1 \rangle}$$

Image of $a \otimes b$ in $K_2(k)$ is denoted $\{a, b\}$

Ex / • $K_2(\text{finite field}) = 0$

e.g. $K_2(\mathbb{F}_3) = 0$ bc $\mathbb{F}_3$ is gen by -1
and $\{-1, -1\} = 0$

$$\bullet K_2(\mathbb{Q}) \stackrel{\text{Tate}}{=} \mathbb{Z}/2\mathbb{Z} \oplus \bigoplus_{\substack{p \text{ odd} \\ \text{prime}}} (\mathbb{Z}/p\mathbb{Z})^\times$$

$$\bullet K_2(\mathbb{R}) = \mathbb{Z}/2\mathbb{Z} \oplus \text{uniquely divisible}$$

$$\bullet K_2(\mathbb{C}) \text{ uniquely divisible}$$

(i.e. $\forall n \in \mathbb{Z}^+, a \in K_2$
 $nb = a$ for some $b \in K_2$)

Historical timeline

1968

Matsumoto's thesis describes $K_2(F)$

1970

Milnor introduces $K_*^M(F)$

Matsumoto's work motivates

$$K_n^M(k) = (k^\times)^{\otimes n} / \langle a_1 \otimes \cdots \otimes a_n : a_i + a_{i+1} = 1 \rangle$$

Milnor's conjecture / Bloch-Kato

We have isom.

$$K_n^M(k) / \ell K_n(k) \xrightarrow{\sim} H_{\text{ét}}^n(\text{Spec } k, \mu_{\ell}^{\otimes n})$$

$\forall n, \ell \geq 0$

motivates Voevodsky to introduce
motivic coh.

Rmk. Milnor K-theory

Quillen K-theory

$$K_n^M(k) = K_n(k) \quad n=0,1,2$$
$$\neq \quad \text{in general}$$

Exercise. $K_n^M(\text{finite field}) = 0, \quad \forall n \geq 2$

(But $K_n(\text{finite field}) \neq 0$
for n odd)

